
The `_ppNR_` Dataset Family, v2

31 grids under one configuration: clean DC start, perturbed static generation, and a PV-bus reactive entry consistent with the stored voltage

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Abstract

This document describes the **v2** `_ppNR_` dataset family: 31 parquet files covering 31 power grids from 4 to 9,241 buses, every one generated under an *identical* configuration for the first time. The v1 family documented previously was internally inconsistent in three ways, each of which is measured and corrected here. (i) The 28 PPC grids carried $5^\circ/0.02$ pu noise on `u_start` while the 3 CGMES grids did not; the noise is now off everywhere, because it degrades `u_start` as a predictor of `u_newton` by 35–134% while adding no information. (ii) `net.sgen` was never perturbed, freezing 56.2% of scheduled generation on case24_ieee_rts, 34.5% on case89pegase, 17.2% on LVN heo1 and 100% on SimBench across all scenarios; static generation now follows the load scale and the dispatch jitter. (iii) The PV-bus reactive entry of `S_start` was back-computed at a flat start voltage that is *not* the `u_start` stored beside it, inconsistently so per grid; it is now recomputed from `u_start`. We also document the per-grid load-scale feasibility cliff that explains every shortfall yield in v1 to within one percentage point, and a systematic DC-versus-AC angle bias that accumulates with distance from the slack.

1 Location and Selection

All files reside in a single directory on the FAU Fritz cluster:

```
/home/vault/iwi5/iwi5295h/PIGNN-Attn-LS/ScenarioSynthesis_PPC/out/
```

The v1 selection rule no longer works. Selecting on the `_ppNR_` solver tag alone now matches several generations of the same grids. The v2 family is selected by its configuration tag:

```
ls .../out/*_u0clean_sgenpert_pvqvstart_*directSI.parquet
```

This yields exactly 31 files. The tag is authoritative: it is emitted by the filename builder only when all three of `-rand_u_start` (absent), `-perturb_sgen` and `-pv_q_from_vstart` hold, so no earlier generation can be selected by accident. Anything matching `_cNR_` is a different family entirely — it was labelled with the in-repo custom solver, not `pandapower` — and is excluded automatically.

2 Filename Scheme

Filenames encode the full experimental configuration so that runs differing in any controlled variable cannot collide:

```
<grid>_<ybus>_<preset>_<start>_<solver>_<load>_  
<u0>[_<flags>]_<unit>_<rows>_NR_branchrows_directSI.parquet
```

For example:

case118_ppcY_backbone_dc_compile_ppNR_ls0.60-1.40_
u0clean_sgenpert_pvqvstart_siNR_37000_NR_branchrows_directSI.parquet

Field	Meaning
ppcY	Y-bus taken from the pandapower PPC (not re-stamped)
backbone	scenario preset (Section 3)
dc_compile	initial voltage from a DC power-flow solve
ppNR	label solver = pandapower Newton–Raphson
ls0.60-1.40	effective global load-scale range ($\pm 40\%$)
u0clean	no start-point jitter (Section 4.1)
sgenpert	net.sgen is perturbed (Section 4.2)
pqvstart	PV-bus Q from u_start (Section 4.3)
siNR	NR solved and stored in SI units (not per-unit)
37000	true row count in the file

The row-count field is now trustworthy. In v1 it was built from `-runs`, which counts *attempts*. With `-drop_nonconverged` the two differ, so `case145_..._36000_...` actually held 7,885 rows and `case300_..._36000_...` held 27,994. Merged array datasets were named by the merger and did carry true counts, so the two halves of the v1 corpus meant different things by the same field. The driver now renames the file to the true retained count on close, and this is verified against the parquet metadata for all 31 files.

3 Perturbation: the backbone Preset

Every file uses `-scenario_level backbone` with no CLI overrides of the load range. This is a *fixed-Y-bus* perturbation family: topology and admittance are untouched, so all rows of a grid share one admittance matrix and the dataset remains `share_grid-compatible`.

Quantity	Value	Effect
Global load scale	[0.60, 1.40]	light $\rightarrow$ heavy $\rightarrow$ near-collapse
Scale gen. with load	yes	keeps slack burden bounded
Scale sgen with load	yes	new in v2 (Section 4.2)
Per-bus load P jitter	$\sigma_P = 0.10$	decorrelates buses
Per-bus load Q jitter	$\sigma_Q = 0.15$	reactive less predictable than active
Generator jitter	$\sigma_G = 0.05$	dispatch variation, <i>gen</i> and <i>sgen</i>
PV voltage setpoint	[0.95, 1.05] pu	realistic setpoint band
Start-point jitter	off	changed in v2 (Section 4.1)
Line outage probability	0.0	topology fixed
Admittance jitter	0.0	Y-bus fixed

Note the ordering, which matters for Section 4.1: every quantity above is applied to the network *before* the DC power flow that produces `u_start`. They perturb the *scenario*, and `u_start` and `u_newton` move together in response.

Correction to the v1 report. The v1 document asserted that “narrowing [the load range] to $\pm 20\%$ or $\pm 10\%$... changed convergence by less than 6%, establishing that solver choice rather than perturbation magnitude governs yield.” That measurement was made with the custom solver. Under pandapower NR the load range governs yield almost completely, as Section 5 shows.

4 What Changed in v2, and Why

4.1 The start-point jitter is off

In v1 the 28 PPC files were generated while `rand_u_start` was read from the preset, which sets it on (5° angle jitter on all non-slack buses, 0.02 pu magnitude jitter on PQ buses). The generator was later changed to make `rand_u_start` CLI-only and default off, so the three CGMES files, generated

afterwards, had a clean DC start. The v1 report flagged this as “the one substantive inconsistency in the family.”

It is not merely an inconsistency; the jitter is harmful. The learning task is $u_{\text{start}} \mapsto u_{\text{newton}}$, and u_{newton} is a function of (S, Y) alone — it does not depend on the initial guess. Since the model already receives S and Y , u_{start} carries no information about the target. Its entire value is being a *cheap approximation of the answer*, which makes corrupting it a pure loss. Measured mean angle error of u_{start} against u_{newton} , with the per-grid bias of Section 6 removed:

Grid	clean	jittered	degradation
case118	1.145°	2.675°	+134%
case300	1.959°	3.216°	+64%
case1354pegase	2.997°	4.041°	+35%

The jitter is also the wrong instrument for start-point diversity. A DC power flow is a linear solve, $\theta = B^{-1}P$; it takes no initial guess, so noise applied *before* it is simply overwritten. What does reach the DC solution is the scenario perturbation of Section 3 — and PV-bus magnitudes already carry it, since the DC start sets $|V| = 1.0$ pu everywhere and then overwrites generator buses with their (jittered) setpoints. Only PQ magnitudes were flat, so the $\pm 2\%$ magnitude jitter was adding noise to a constant, and the $\pm 5^\circ$ angle jitter was corrupting the one genuinely informative component.

Verification. Because a clean DC start leaves every PQ bus at exactly 1.0 pu, the absence of jitter is an exact check rather than a threshold. Across all 31 files the maximum deviation of $|u_{\text{start}}|$ from 1.0 pu at PQ buses is $\sim 10^{-16}$, i.e. machine epsilon.

An A/B pair is retained. case118 and case300 exist in both arms — clean and jittered, identical in every other respect — so the claim above can be tested on a trained model rather than argued. These four files are kept outside the v2 family and are not selected by the tag in Section 1.

4.2 Static generation is perturbed

`net.sgen` holds static generators: wind, PV plants, CHP — PQ injections with a specified p and q . In v1 the perturbation code never referenced `net.sgen` at all. The global load scale and the dispatch jitter reached `net.gen` and `net.load` only, so *static generation was identical in all 36,000 scenarios of every grid*.

This is negligible where `sgen` is negligible and decisive where it is not:

Grid	sgen units	sgen MW	share of scheduled generation
SimBench	121	3,715.6	100.0%
case24_ieee_rts	22	1,632.2	56.2%
case89pegase	6	2,430.8	34.5%
case3120sp	257	5,235.7	25.1%
case6495rte	1,157	19,139.4	17.9%
case6470rte	1,125	19,428.9	17.5%
case6515rte	1,158	20,810.2	17.5%
LVN heol	33	36.9	17.2%
case5	1	170.0	17.0%
case9241pegase	434	23,055.8	6.8%
case2848rte	300	2,743.1	5.0%
case2869pegase	180	6,497.6	4.7%
case1354pegase	52	1,086.3	1.5%
case300	8	321.8	1.4%
case1888rte	82	679.9	1.1%

Thirteen of 28 PPC grids and 2 of 3 CGMES grids exceed 1%. **SimBench is the extreme case:** with `net.gen` empty and all 3,716 MW in `sgen`, sweeping the load scale from 0.60 to 1.40 moved its total injection from 3,669.9 to 3,588.2 MW — a 2.2% swing where the preset intends $\pm 40\%$. Its 36,000 v1 rows are close to duplicates of one another, which is also why it converged 100% of the time.

Semantics. Static generation now follows the same global scale as `net.gen`, with q scaled alongside p so the unit’s power factor is preserved, and receives the same per-unit dispatch jitter with p and q drawn independently (matching how loads are treated). Unlike p , q is signed and is not clamped: an `sgen` may legitimately absorb reactive power.

Verification. Comparing injection summed over `sgen` buses across rows, v1 versus v2:

Grid	v1 spread	v2 spread
case89pegase	0.0%	78.7%
case2848rte	9.7%	79.7%
case6470rte	15.2%	55.1%
case24_ieee_rts	30.9%	86.2%

case89pegase is the cleanest demonstration: its `sgen` buses carry no load, so in v1 their injection was constant to the digit (2430.8–2430.8 MW). The v2 spread of $\sim 80\%$ is what a $[0.60, 1.40]$ scale implies.

Effect on yield. Measured at $n=60$ per grid, perturbing `sgen` moves convergence by less than the sampling noise on every PPC grid ($\pm 5\%$, mostly 0%). SimBench is the exception, falling from 100% to 85–95% — which is not a regression but the grid finally having an operating point that moves.

4.3 The PV-bus reactive entry now matches `u_start`

At a PV bus, Q is not a specified quantity, and what `S_start` carried in v1 was not a placeholder either. The pipeline compiles the PPC with

```
pp.runpp(net, init="flat", max_iteration=1, tolerance_mva=1e9)
```

whose enormous tolerance makes it converge immediately, after which `pypower`’s `pfsoln` back-computes each generator’s Q_g from whatever voltage the PPC then holds — V_0 : flat magnitudes with the generator setpoints applied, zero angles. `makeSbus` folds that into `s_multi`. Verified to machine precision on case118, case300 and case1354pegase:

$$\text{s_multi.imag[PV]} = \text{Im}[V_0 \overline{\mathbf{Y}V_0}] \quad (10^{-16} \text{ to } 10^{-13} \text{ pu}).$$

The problem is that V_0 is *not* `u_start`. With `start_mode=dc_compile`, `u_start` carries DC angles, so the stored reactive entry describes a different operating point from the voltage shipped beside it — and inconsistently per grid: the correlation between `s_multi.imag[PV]` and $Q(\text{u_start})$ is 0.911 on case118 but 0.568 on case300. A consumer of the data cannot see this.

In v2 the entry is recomputed as

$$Q_{\text{start}}[\text{PV}] = \text{Im}[u_{\text{start}} \overline{\mathbf{Y} u_{\text{start}}}],$$

so the column means the same thing on every grid: *the reactive injection this network implies at the initial voltage you were given*. Active power at PV buses is untouched — P there is specified. This uses only `u_start` and $\mathbf{Y}$, never `u_newton`, so it is not label leakage.

Verification. Rebuilding $\mathbf{Y}$ from the stored branch columns with the dataloader’s own routine and evaluating the identity gives a maximum relative deviation of $\sim 10^{-14}$ across all files with PV buses. Grids with no PV buses at all — SimBench has 3 slack and 91 PQ, since all its generation is `sgen` — are unaffected by this change; their fix is Section 4.2.

5 Why Some Grids Retain Fewer Rows

Every grid has a sharp *global load-scale threshold* above which the AC power flow does not converge: 100% below, 0% above, with no gradual falloff. Measured by sweeping the load scale with all jitter disabled:

Grid	last 100% scale	first 0%	$P(\text{scale} < \text{cliff})$	v1 yield
case6515rte	1.00	1.05	56.3%	55.9%
case9241pegase	1.00	1.05	53.1%	50.8%
case145	1.05	1.10	59.4%	21.9%
case6495rte	1.10	1.15	65.6%	64.8%
case6470rte	1.15	1.20	71.9%	71.0%
case300	1.30	1.40	93.8%	77.8%
case1354pegase	1.30	1.40	93.8%	90.6%
case1888rte	> 1.40	—	100%	100%
case2869pegase	> 1.40	—	100%	100%

The one-parameter model “yield = probability the load scale lands below the cliff” reproduces five of the seven shortfall yields to within one percentage point. This is genuine infeasibility, not solver fragility.

Two grids carry a second mechanism. **case145** sits on its own cliff, so per-bus load jitter alone pushes samples across it: at scale 1.00 it converges 100% of the time with no jitter but 63/30/17% at $\sigma_P = 0.05/0.10/0.15$. **case9241pegase** is sensitive to the PV *setpoint*: at scale 1.00, a fixed 1.0 pu setpoint converges 100% of the time while the preset’s $U(0.95, 1.05)$ gives 53%; per-bus load jitter is irrelevant there.

A mechanism that was *ruled out*: per-bus Q jitter flipping a load’s sign. That requires $z < -1/\sigma_Q = -6.67$, probability 1.3×10^{-11} ; the measured rate is 0.000 on every grid. Per-bus jitter also averages out in aggregate, so large grids see *less* total load variation, not more — bus count alone is not the driver.

Consequence for v2. Because the boundary is sharp in load scale, the retained rows of a shortfall grid are already samples from $U(0.60, \text{cliff})$ rather than $U(0.60, 1.40)$; oversampling and narrowing the range give nearly the same data. v2 oversamples so that all 31 grids keep one identical preset, with `-runs` set to 37,000/(measured yield).

6 A Systematic DC-versus-AC Angle Bias

`u_start` and `u_newton` agree exactly at the slack (0.00°) but differ at every other bus by a large, grid-specific amount: a mean of 2.05° on case118, 18.57° on case300 and 16.77° on case1888rte. Removing that offset leaves a residual of only $1\text{--}2^\circ$.

This is not a gauge artifact. A global phase rotation would preserve every angle *difference*; here the differences themselves change (up to 9.87° on individual branches). It is the intrinsic modelling error of DC power flow, integrated along the network. DC assumes $|V| = 1$ pu, $\sin \theta \approx \theta$ and $R = 0$; the lossless assumption is directly visible in the slack injection, which is 47.7 MW in DC against 472.4 MW in AC on case300 — the entire 424.7 MW of network losses is missing. Every branch’s angle drop is therefore slightly wrong, *with a consistent sign*, and bus angles are sums of branch drops along a path from the slack. Measured against hop distance:

hops from slack	case300 mean d	case118 mean d
0 (slack)	0.00°	0.00°
1	2.74°	1.10°
2	11.07°	1.46°
4	14.96°	2.29°
8	17.03°	3.91°
12	19.93°	—

$\text{corr}(\text{hop}, d) = 0.742$ on case300 and 0.695 on case118. Mean per-branch error is only 0.442° on case300; twelve hops of it accumulate to 19.93° . case300 is worst because its slack is a stub with a single incident branch, its diameter is 17 hops, and it spans 13 voltage levels down to 0.6 kV, where $R \sim X$ and the DC assumption fails hardest.

Implication. The bias is physically real and cannot be normalised away; u_{start} and u_{newton} are each correct states of their own model. But it means the model must learn a grid-specific, distance-dependent correction between its input and its target, and the magnitude of that correction varies with grid diameter, voltage spread and loss fraction — which makes it a genuine obstacle to cross-grid transfer. It also masks smaller effects: the start-jitter degradation in Section 4.1 is invisible until this offset is removed.

7 Generation Procedure

All 31 files come from `main_datagen_multiproc_improved.py` with `-label_solver pandapower_nr`, `-start_mode dc_compile`, `-K 40`, `-convergence_mode misinf`, `-drop_nonconverged`, `-no_save_y_matrix`, `-perturb_sgen` and `-pv_q_from_vstart`, and *without* `-rand_u_start`. The 28 PPC grids additionally pass `-use_force_shunt_when_no_trafo`.

Unlike v1, every grid ran as a **single-node job** rather than a chunked array with a merge step. Chunked runs need the chunk files and the merged file on disk simultaneously, and the vault filesystem was at its soft quota; measured throughput made one node sufficient for all 31.

The three CGMES grids pass `-cgmes_path`, `-cgmes_version 2.4.15` and `-cgmes_ignore_errors`. SimBench and ENTSO-E use `-no_cgmes_model_a_cleanup`; LVN heol uses `-cgmes_model_a_cleanup`, `-use_force_shunt_when_no_trafo` and `-base_sn_mva 100`. The last is mandatory: every LVN CGMES export carries `net.sn_mva = 1`, at which the specified injections reach $\sim 162,000$ pu and pandapower’s NR cannot reach its tolerance at any iteration count.

Every sample is attempted independently: a fresh perturbation is drawn, a DC power flow provides the initial voltage, pandapower NR solves the AC problem, and the row is written only if it converged. **Non-converged samples are discarded, not stored**, which is why `-runs` exceeds the retained count wherever the grid has a feasibility cliff.

8 Row Schema

Each row is one complete power-flow scenario: 30 columns, ZSTD-compressed. Array-valued fields are stored as serialised NumPy buffers (binary), so a row is self-contained and the grid topology is repeated per row rather than normalised out.

Group	Columns	Type
Sizes	<code>bus_number</code> , <code>branch_number</code>	int32
Base	<code>gridtype</code> , <code>U_base</code> , <code>S_base</code> , <code>vn_kv</code>	string/double
Bus	<code>bus_typ</code> , <code>Y_shunt_bus</code>	binary
Branch	<code>Branch_f_bus</code> , <code>Branch_t_bus</code> , <code>Branch_status</code> , <code>Branch_tau</code> , <code>Branch_shift_deg</code> , <code>Is_trafo</code> , <code>Branch_hv_is_f</code> , <code>Branch_n</code>	binary
Admittance	<code>Branch_y_series_from/_to/_ft</code> , <code>Branch_y_shunt_from/_to</code> , <code>Y_Lines</code> , <code>Y_C_Lines</code>	binary
Input	<code>u_start</code> , <code>S_start</code>	binary
Label	<code>u_newton</code> , <code>S_newton</code>	binary
Diag.	<code>converged</code> , <code>final_misinf</code> , <code>nr_iterations</code>	int8/double/int32

The learning task is $u_{\text{start}} \mapsto u_{\text{newton}}$. Only the converged *endpoint* is stored, not the NR iterate trajectory. `nr_iterations` is 0 throughout — the pandapower label path does not report an iteration count — so it carries no information here. `Y_matrix` is absent by design; the admittance is reconstructed from the branch columns at load time.

S_start is not load-only. It is the signed injection before the solve: perturbed load everywhere, plus each unit’s scheduled dispatch at controllable buses, plus — at PV buses — the reactive entry of Section 4.3. At non-controllable buses S_start equals S_newton to $\sim 10^{-6}$ MW; at slack and PV buses they differ by construction, because slack injection and PV reactive power are solver outputs.

9 Contents

Table 1 lists every file. *Attempts* is `-runs`; *yield* is retained rows over attempts.

Table 1: The 31 v2 _ppNR_ datasets, ordered by network size.

Grid	Buses	Br.	Rows	Att.	Yield	MB	kV lv.	kV range	Src
case4gs	4	4	37,000	37,000	100.0%	8.2	1	230	PPC
case5	5	6	37,000	37,000	100.0%	11.0	1	230	PPC
case6ww	6	11	37,000	37,000	100.0%	13.2	1	230	PPC
case9	9	9	37,000	37,000	100.0%	18.0	1	345	PPC
case14	14	20	37,000	37,000	100.0%	30.8	4	0.208–135	PPC
case24_ieee_rts	24	38	37,000	37,000	100.0%	52.4	2	138–230	PPC
GBReducednetwork	29	99	37,000	37,000	100.0%	65.6	3	132–400	PPC
case30	30	41	37,000	37,000	100.0%	63.5	1	135	PPC
case_ieee30	30	41	37,000	37,000	100.0%	64.2	4	1–132	PPC
case33bw	33	32	37,000	37,000	100.0%	74.0	1	12.66	PPC
case39	39	46	37,000	37,000	100.0%	82.2	1	345	PPC
case57	57	80	37,034	37,100	99.8%	122.0	6	115–500	PPC
case89pegase	89	210	36,999	37,000	100.0%	175.9	3	150–380	PPC
SimBench	94	131	38,649	45,000	85.9%	188.7	3	110–380	CGMES
case118	118	186	37,000	37,000	100.0%	263.1	3	138–345	PPC
case145	145	453	36,918	170,000	21.7%	290.5	7	18–500	PPC
iceland	189	206	37,112	37,500	99.0%	361.7	15	0.22–220	PPC
case_illinois200	200	245	37,000	37,000	100.0%	424.1	3	13.8–230	PPC
case300	300	411	37,930	48,000	79.0%	659.4	13	0.6–345	PPC
LVN_heo1	722	796	37,000	37,000	100.0%	1,310.7	8	3–380	CGMES
case1354pegase	1,354	1,991	37,038	41,000	90.3%	2,785.6	2	220–380	PPC
case1888rte	1,888	2,531	37,000	37,000	100.0%	3,943.5	20	3–380	PPC
GBnetwork	2,224	3,207	37,022	38,800	95.4%	4,162.6	8	6.6–400	PPC
case2848rte	2,848	3,776	37,000	37,000	100.0%	5,906.2	20	3–380	PPC
case2869pegase	2,869	4,582	37,099	37,100	100.0%	6,027.2	4	110–380	PPC
case3120sp	3,120	3,693	37,191	37,600	98.9%	6,616.5	6	16–400	PPC
ENTSO_E_RealGridTest	6,051	8,202	37,000	37,000	100.0%	12,475.1	8	17–380	CGMES
case6470rte	6,470	9,005	37,895	52,000	72.9%	13,779.4	19	3–380	PPC
case6495rte	6,495	9,019	37,889	58,000	65.3%	13,844.3	21	3–380	PPC
case6515rte	6,515	9,037	37,211	67,000	55.5%	13,621.2	21	3–380	PPC
case9241pegase	9,241	16,049	38,508	75,000	51.3%	20,589.8	9	110–750	PPC
Total (31)			1,153,495	1,410,100	81.8%	108,030.5			

10 Verification

All 31 files were checked directly against the stored bytes, not against the flags that were passed:

- **Row count** matches the filename field and the parquet metadata; every grid retains at least 36,900 rows.
- **converged** = 1 for every row of every file.
- **Loader columns** `u_start`, `u_newton`, `S_start`, `S_newton` present in all 31.
- **Clean start:** $\max ||u_{\text{start}}|/V_{\text{base}} - 1|$ over PQ buses is $\sim 10^{-16}$.
- **PV-bus Q:** $\text{Im}[u_{\text{start}} \overline{\mathbf{Y} u_{\text{start}}}]$ reproduced to $\sim 10^{-14}$ relative, with $\mathbf{Y}$ rebuilt from the stored branch columns.
- **sgen perturbation:** injection summed over sgen buses spans $\sim 80\%$ of its mean, against 0.0–30.9% in v1.

11 Known Caveats

- **The DC-versus-AC angle bias of Section 6 remains**, because it is physically real. It is grid-specific and distance-dependent, and is the largest single obstacle to cross-grid transfer in this corpus.
- **Yields are not uniform.** Grids with a low feasibility cliff spend many attempts per retained row, and their retained rows sample $U(0.60, \text{cliff})$, not $U(0.60, 1.40)$. The load-scale marginal therefore differs per grid even though the *attempt* distribution is identical.
- **final_misinf is written at sampling time** from the in-memory Y-bus against the specified injection vector, not recomputed from the stored file, and it is in SI units (VA) — read it relative to S_{base} .
- **LVN heo1 carries asymmetric branch impedance** (BR_R_ASYM, BR_X_ASYM), so $Y_{ft} \neq Y_{tf}$. Consumers building their own Y must derive each direction from the from/to admittances scaled by the voltage-base ratio rather than reusing `Branch_y_series_ft` for both; the naive reconstruction gives a 9% relative residual on LVN against 10^{-9} when done correctly.

12 Summary

- **31 files, 1,153,495 rows, 108.0 GB**, selected by the `u0clean_sgenpert_pvqvstart` tag. 1,410,100 scenarios were attempted for an overall yield of 81.8%, and *every* grid retains at least 36,918 rows — against `v1`, where six grids fell between 18k and 33k. This is the first internally consistent generation of this corpus.
- **Clean DC start** everywhere. The removed jitter degraded `u_start` by 35–134% as a predictor of `u_newton` while adding no information; a retained A/B pair on `case118` and `case300` lets this be tested rather than assumed.
- **Static generation is perturbed**, fixing a defect that froze 100% of SimBench’s generation, 56% of `case24_ieee_rts`’s and 17% of LVN `heo1`’s across all scenarios.
- **PV-bus Q is consistent with the stored voltage**, so the column means the same thing on every grid.
- **Row counts in filenames are true counts**, verified against parquet metadata — they were attempt counts on the single-node half of `v1`.
- **Shortfall yields are explained**: each grid has a sharp load-scale feasibility cliff, and a one-parameter model built from it reproduces five of seven `v1` yields to within one percentage point.